Andrew and Erna Viterbi Faculty of Electrical Engineering
Electronics
Computers
Communications

VLSI Laboratory המעבדה ל-

AdQuanta

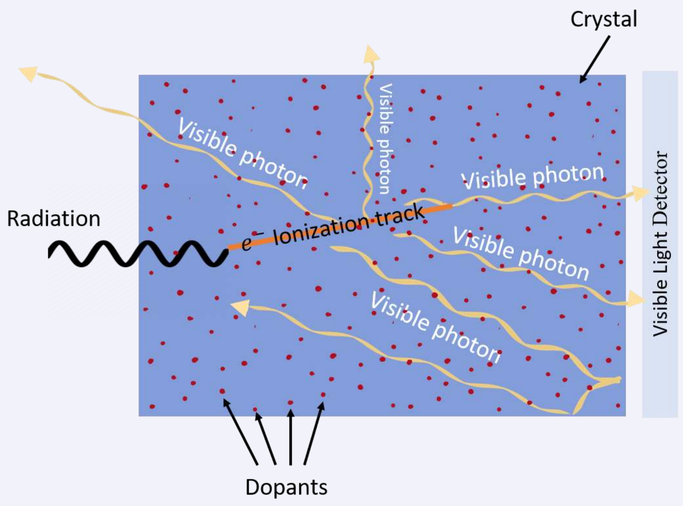
TECHNION
Israel Institute of Technology

Optimizing the Design of Radiation Detectors Using Nanostructures

Ohad Segal and Avner Shultzman - Supervised by Yaniv Kurman

Scintillator Based Radiation Detectors

- High energy photon incident on a scintillator bulk generates a single electron
- The electron creates e-h pairs that recombine in dopants to create visible photons

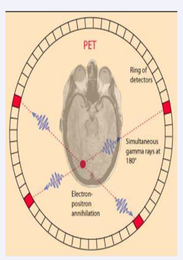


Project Goal

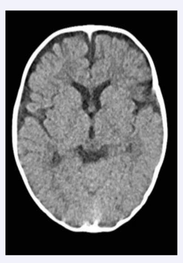
Develop a new tool for the design and optimization of novel high energy radiation detectors (Scint City)

Applications

PET Scans



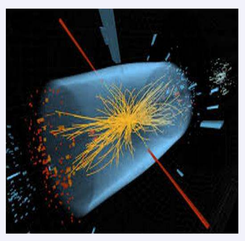
CT Scans



Mammography and X-ray Imaging



High Energy Physics CERN



Night Vision



Theoretical Basis

- First an expression for spontaneous emission rate for multi-layers nanostructures was derived
- The emission rate of a single dopant is given by :

$$\sum_{\sigma}\Gamma(\omega,\theta,t,\sigma,\vec{n},\vec{d})=\frac{k_3 l_3^2}{k_1^3}\left(k_1^2\frac{\varepsilon_1}{\varepsilon_3}\left|\frac{T_{\parallel s\uparrow}}{l_1}\exp(il_1t)\right|^2+\left|T_{\parallel p\uparrow}\exp(il_1t)\right|^2+u^2\left|\frac{T_{\perp p\uparrow}}{l_1}\exp(il_1t)\right|^2\right)$$

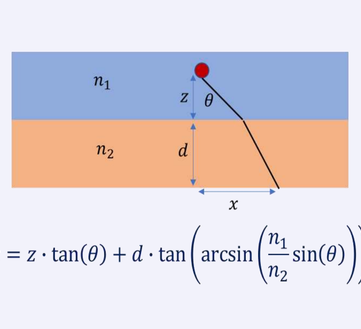
$$T_{\parallel s\backslash p\uparrow}=\frac{t_{13,s\backslash p}\left(1+r_{12,s\backslash p}e^{2i(d-t)}\right)}{1-r_{12,s\backslash p}r_{13,s\backslash p}e^{2il_1d}},\ u=k_{max}\sin\theta,\ l_i=\sqrt{k_i^2-u^2}$$

- Integrate to obtain the total effective emission of a structure

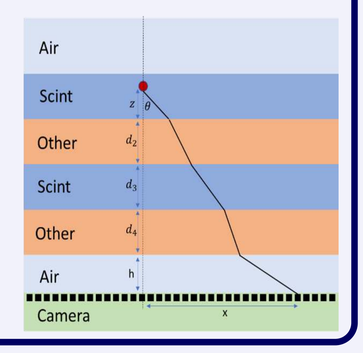
$$\Gamma_{\mathrm{eff}}(\theta,\vec{n},\vec{d})=\int d\omega Y(\omega)\int_0^L\frac{dz}{L}\sum_{\sigma=\{TE,TM\}}\Gamma(\omega,\theta,z,\sigma,\vec{n},\vec{d})$$

Processing Images with Multilayers Nanostructures

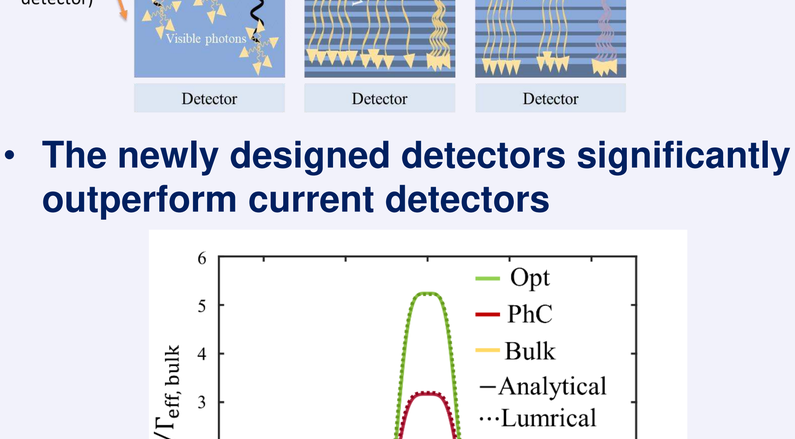
- Analytical calculation of the impulse response of multi-layers nanostructures
- Use Purcell effect and Snell's law
- **A brand new visualization technique!**



$$x=z\cdot\tan(\theta)+d\cdot\tan\left(\arcsin\left(\frac{n_1}{n_2}\sin(\theta)\right)\right)$$

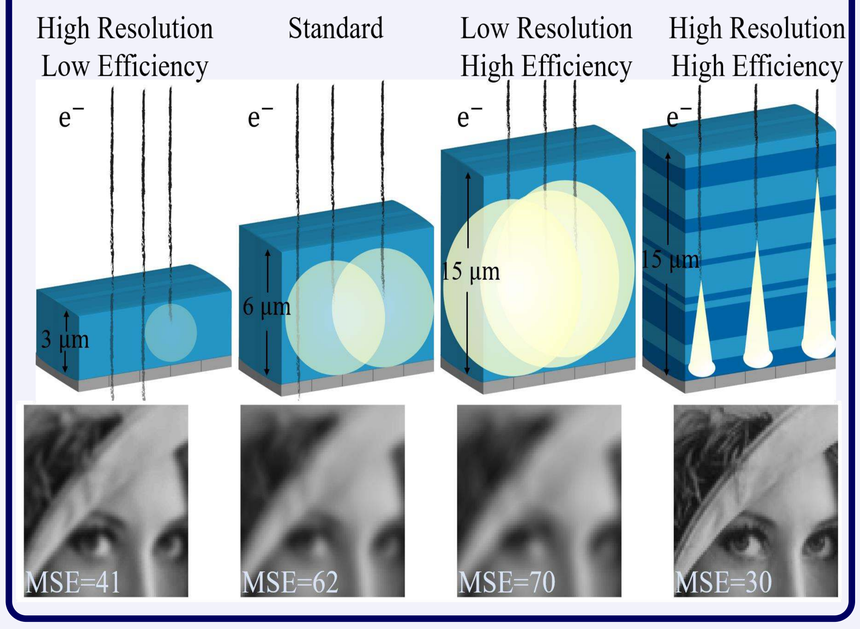


Multidisciplinary Engineering



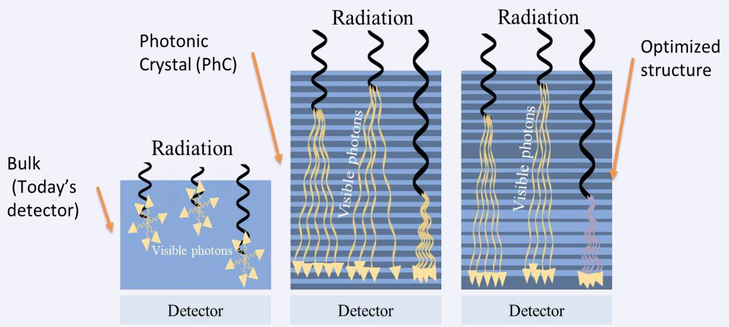
Thick or Thin Bulk
Getting the Best of Both Worlds

- The novel nano-structures overcome existing efficiency-resolution trade-offs

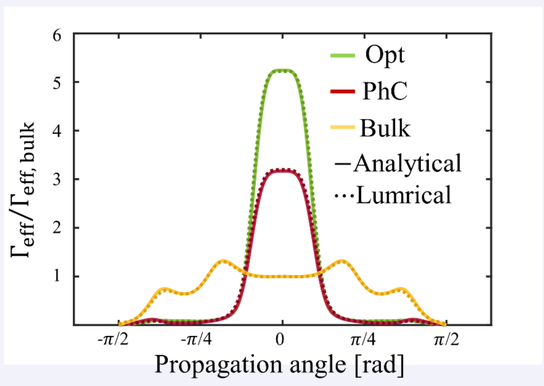


Optimizing Nanostructures for X-ray Detection

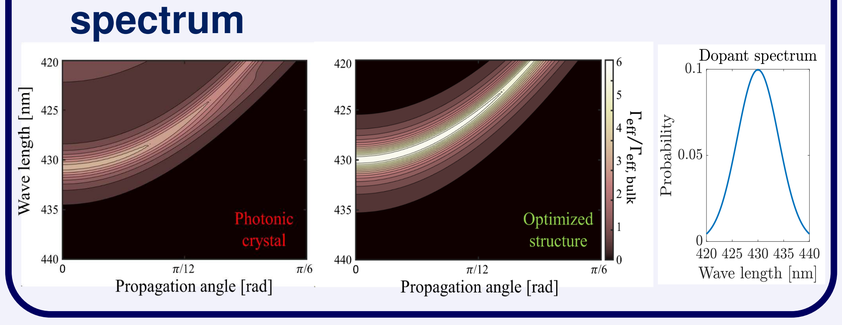
- Two detectors for X-ray imaging were designed with the new technique



- The newly designed detectors significantly outperform current detectors

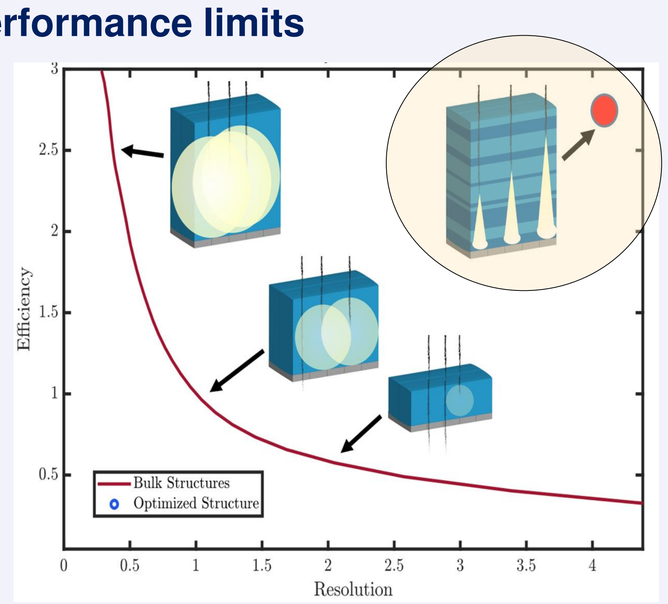


- Peak of the structures' enhancement match the peak of the emitters' spectrum



Overcoming Traditional Detector Design Trade-offs

- Existing designs follow a standard efficiency-resolution curve
- Resolution always traded-off for efficiency and vice-versa
- Nanostructures shatter the current performance limits

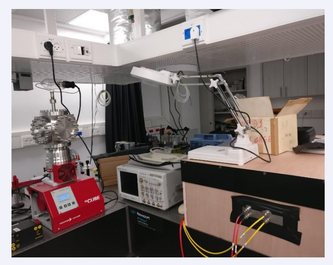


$$\text{Efficiency}=\frac{\int \text{PSF}(x)dx}{\int \text{Standard_PSF}(x)dx}$$

$$\text{Resolution}=\frac{\text{Cut off}(\text{FFT}\{\text{PSF}\})}{\text{Cut off}(\text{FFT}\{\text{Standard_PSF}\})}$$

Immediate Impact

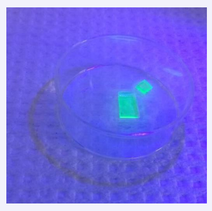
- New scintillator lab in the Solid-State Institute



- X-ray detectors based on new design techniques manufactured and tested in CERN



- Design of efficient UV sources for harmless disinfection



Conclusions

- **Nano-structures** can implement **novel** High Energy Radiation (HER) **detectors** that **overcome** the **traditional** resolution-efficiency **trade-off**
- A **multidisciplinary** project, to achieve **breakthroughs** in HER detector design
- **Scint City** – our new tool, capable of designing **optimized** HER detectors for a **wide range** of applications